

Superposition Without Time

What the Absence of the Time Operator in QM Means
and How $\tilde{T} \cdot m = 1$ Resolves It Structurally

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Preliminary note

This document builds on Doc. 307 (time in the state space or beside it) and Doc. 333 (superposition in FFGFT), without repeating their content. The specific focus here: what does it mean concretely for the interpretation of *superposition, entanglement and collapse* that standard QM has no time operator? And what changes structurally when $\tilde{T} \cdot m = 1$ gives every mode its own time scale?

.1 The Classical Background — Instantaneity as an Inheritance

Classical mechanics: time runs, calculations are instantaneous

Instantaneity is not a QM-specific problem. It is an inheritance from classical physics that QM took over unchanged.

In Newtonian mechanics, gravity acts instantaneously over arbitrary distances. The Coulomb potential is instantaneous. Lagrangian and Hamiltonian mechanics describe the state $(q(t), p(t))$ at a fixed t — the calculation is a timeless algebraic evaluation over the instantaneous state. Nobody asks how long this calculation “takes”, because calculations in this language have no duration. The time t runs in the background as an external parameter — the physics resides in the timeless structure of the equations.

This is the same structure as in QM. Pauli only made explicit what was tacit in the classical framework.

The hidden inconsistency of QM

QM inherits this instantaneity — and adds a new tension:

The state $\psi(t)$ is instantaneous. It is a timeless mathematical quantity at a fixed t — no duration, no process.

Evaluating $|\psi(t)|^2$ requires time. $|\psi(t)|^2$ is a probability density — to verify it one needs many measurements, hence real physical time. The statistics that confirm the state are distributed over a temporal process. The formalism, however, describes only the state at one moment.

This is the tacit inconsistency: *the formalism is instantaneous, but its verification is temporally extended*. The time required for probabilistic evaluation stands outside the formalism — it is neither operator nor variable, but what the experimenter expends in order to know the state at all.

Consequence for superposition

Superposition $c_1 |\psi_1\rangle + c_2 |\psi_2\rangle$ is therefore doubly timeless: first as a state (no time operator, no intrinsic “when”), second as a statement (probabilistic, requires time for verification, which lies outside the formalism). What “exists” in superposition can only be confirmed through repeated measurements over real time — and each individual measurement collapses the state, thus destroying the superposition.

In FFGFT: the period $\tilde{T}_i = 1/m_i$ gives the mode an intrinsic time scale. Superposition is not timeless — it has a geometrically defined scale on which it is stable or unstable.

.2 The Absence of Time in QM — an Underestimated Consequence

The Pauli theorem and its consequences

Wolfgang Pauli showed in 1933 that for a Hamiltonian with a spectrum bounded from below there can be no self-adjoint time operator canonically conjugate to $\hat{H}$ (Doc. 307, Doc. 306). The consequence was tacit but far-reaching: time was kept outside the Hilbert space and treated as a classical external parameter.

This is not merely a technical decision. It means: **time is not an observable in QM**. One cannot ask “when is the system in state ψ ?” and obtain an answer from the formalism — t must be supplied from outside.

What is rarely stated in textbooks

Most textbooks introduce time as a parameter without emphasising what the absence of a time operator means. It means concretely:

- Superposition $c_1 |\psi_1\rangle + c_2 |\psi_2\rangle$ is a state *without time structure* — there is no intrinsic “clock” saying how long the system remains in this superposition.
- Entanglement between two particles is *timeless* — the joint state has no built-in notion of simultaneity.
- Collapse is *without duration* — it “happens” in the external parameter t , but t itself is not part of the collapse process.

This is not a failure of QM — it is a structural property that is rarely made explicit.

.3 Three Consequences in Detail

Superposition is timeless

An electron in the state $|\psi\rangle = c_\uparrow |\uparrow\rangle + c_\downarrow |\downarrow\rangle$ has no intrinsic time scale. The question “how long is it in this superposition?” is not answerable within the QM formalism — one would need a

time operator, which does not exist. What does exist is the coherence time T_2 — but that is a phenomenological quantity describing interaction with the environment, not an intrinsic property of the state.

In FFGFT this is different: every mode $\mathcal{T}_i = (n_i, l_i, j_i)$ carries $\tilde{T}_i = 1/m_i$ as a geometric quantity. A state *is* its time scale — not as the duration of a superposition, but as a structural property of the mode.

Entanglement without simultaneity

Two entangled particles share a joint state $|\Psi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$. This joint state has no built-in time structure. The question of the simultaneity of a measurement on both particles is therefore not a QM question but a classical question about the external parameter t — and hence a relativistic question (relativity of simultaneity).

This is the deepest reason why QM is compatible with Lorentz invariance: because “simultaneity” stands outside the formalism and can therefore be transformed by the Lorentz group without touching the QM formalism.

In FFGFT: if two different modes with $m_1 \neq m_2$ were entangled, they would carry different $\tilde{T}_1 \neq \tilde{T}_2$. A cross-mode entanglement therefore generates incompatible time scales — structurally more problematic than in QM, where this conflict does not arise because t stands outside.

Collapse without duration

The measurement process in QM describes a transition $|\psi\rangle \rightarrow |\psi_k\rangle$ (projection onto an eigenstate). This transition has *no duration* in the QM formalism — it is instantaneous in the parameter t . How long the collapse takes is a question that lies outside standard QM (it is discussed in theories of the measurement process, decoherence, or quantum gravity).

This is not an inconsistency but a direct consequence of the absence of the time operator: without $\hat{t}$ there is no observable “collapse duration”.

In FFGFT the type-I step (measurement act, Doc. 285/333) is not instantaneous in the abstract sense but a geometric transition — the mass circle S_m^1 becomes the time axis $\mathbb{R}_t$, the winding number is discarded. This is a loss of information that has a natural “direction”.

.4 What $\tilde{T} \cdot m = 1$ Changes Structurally

Time as a property of the mode, not a parameter

The fundamental relation $\tilde{T} \cdot m = 1$ (Doc. 285, 306, 307) sets time not as an external parameter but as a geometric quantity of every mode: the period $R_m = \hbar/(mc^2)$ is intrinsic — it belongs to the particle as its mass does.

This has three direct consequences for the points above:

Superposition: A superposition within one mode — intermediate torsion (Doc. 174) — has a natural time scale $\tilde{T}_i$. Superposition *across* different modes would overlay incompatible time scales $\tilde{T}_i \neq \tilde{T}_j$ — this is the structural reason why cross-mode superpositions are problematic in FFGFT (Doc. 333).

Entanglement: Within one mode, simultaneity is defined by $\tilde{T}$. Across modes there is no shared time concept — similar to QM, but for a different reason: not because t stands outside, but because different modes carry different geometric time scales.

Collapse (Type I): The measurement act (Doc. 285) has a natural direction — lossy from S_m^1 to $\mathbb{R}_t$ — and a natural asymmetry: the reverse direction $\mathbb{R}_t \rightarrow S_m^1$ requires the discarded

winding number. The arrow of time is thereby not an external additional assumption but a consequence of the geometry of measurement.

Pauli theorem and compact time

The Pauli theorem applies to an *open* time parameter conjugate to a Hamiltonian bounded from below. For a *compact* time dimension the argument does not apply (Doc. 307): compact time behaves like an angle variable, its conjugate has a discrete spectrum (winding numbers), and the Pauli condition is not met.

$\tilde{T} \cdot m = 1$ realises exactly this situation: time is compact, the conjugate spectrum is the discrete mass ladder. The theorem does not exclude this case — it virtually forces it as the only consistent way to integrate time into the Hilbert space.

.5 Summary and Demarcation

	Standard QM	FFGFT ($\tilde{T} \cdot m = 1$)
Time	absent: no operator, no Hilbert-space element; external parameter	geometric: $\tilde{T}_i = 1/m_i$, compact, mode-bound
Superposition	timeless, no intrinsic clock	intermediate torsion within one mode; $\tilde{T}_i$ as scale
Cross-mode	algebraically permitted, no time conflict (t external)	incompatible $\tilde{T}_i \neq \tilde{T}_j$, structurally problematic
Entanglement	timeless, simultaneity external (Lorentz)	mode-specific $\tilde{T}_i$; cross-mode: no shared time concept
Collapse	instantaneous in t , no duration in formalism	Type-I: directed information loss, arrow of time geometric
Pauli theorem	blocks time operator for open t	does not apply for compact $\tilde{T}$ (angle variable)

This document does not claim to replace standard QM. It shows that its structural property — time as absent — is rarely made explicit, and that $\tilde{T} \cdot m = 1$ offers a geometric alternative that describes the same phenomena (superposition, entanglement, collapse) from a different basic structure.

References: Doc. 174 (superposition as torsion), Doc. 285 (type-I measurement act), Doc. 306 (native time–energy reciprocity), Doc. 307 (time in the state space or beside it), Doc. 333 (mode specificity and superposition).